

Load the packages

```
using DataFramesMeta
using CSV
using ReadStatTables
using Statistics
using CairoMakie
using AlgebraOfGraphics
using PairPlots
using ColorSchemes
using StyledStrings
using SummaryTables
using PairPlots
include(joinpath(@__DIR__, "table-one-mod.jl"))
include(joinpath(@__DIR__, "pairplot-with-regression.jl"))
include(joinpath(@__DIR__, "boxplot-with-stats.jl"))
include(joinpath(@__DIR__, "coerce_string_columns.jl"))
include(joinpath(@__DIR__, "eta-corr.jl"))
include(joinpath(@__DIR__, "gof.jl"))
include(joinpath(@__DIR__, "indiv-fits.jl"))
include(joinpath(@__DIR__, "model-comparison-table.jl"))
# include(joinpath(@__DIR__, "ParameterTables.jl"))
include(joinpath(@__DIR__, "plot-fun.jl"))
include(joinpath(@__DIR__, "session-info.jl"))
```

```
Main.Notebook.sessioninfo
```

Setting Plots theme

```
my_theme = Theme(
  Axis=(
    yminorticks=IntervalsBetween(2),
    yminorgridvisible=true,
    xminorticks=IntervalsBetween(2),
    xminorgridvisible=true,
    xlabelfont=:bold,
    ylabelfont=:bold,
    leftspinecolor=:gray60,
    rightspinecolor=:gray60,
    topspinecolor=:gray60,
    bottomspinecolor=:gray60,
  )
)

my_log_theme = Theme(
  Axis=(
    yminorgridvisible=true,
    xminorticks=IntervalsBetween(2),
```

```

    xminorgridvisible=true,
    xlabelfont=:bold,
    ylabelfont=:bold,
    leftspinecolor=:gray60,
    rightspinecolor=:gray60,
    topspinecolor=:gray60,
    bottomspinecolor=:gray60,
    yscale=log10,
    yminor ticks=IntervalsBetween(10),
    ytickformat=x -> string.(round.(x, digits=1)),
  )
)

set_theme!(my_theme)

```

Load the data

```

path = joinpath(@__DIR__ , "..", "..", "datasets", "derived")
prot002 = CSV.read(
  joinpath(path, "prot002-adppk.csv"),
  DataFrame
)
# prot002 = filter(:COHORTC => !=("Placebo"), prot002)
first(prot002, 5)

```

```

5×63 DataFrame
 Row | STUDYID      USUBJID      RFSTDTC      ARM
...
   | String15     String31      Date      String31
...
 1 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
...
 2 | PROTOCOL-002 PROTOCOL-002-001-00238 2020-02-11 TEST DRUG 200 mg BID
 3 | PROTOCOL-002 PROTOCOL-002-001-00239 2020-02-17 TEST DRUG 200 mg BID
 4 | PROTOCOL-002 PROTOCOL-002-001-00240 2020-02-07 TEST DRUG 50 mg BID
 5 | PROTOCOL-002 PROTOCOL-002-001-00241 2020-02-17 TEST DRUG 100 mg BID
...
                                     59 columns
omitted

```

Dataset to be inspected interactively, to ensure consistency with the expected structure and content.

```
println("No. of rows in the DataFrame: ", nrow(prot002))
println("No. of unique study participants: ", length(unique(prot002.USUBJID)))
println("Total No. of Observations: ", count(==(0), prot002.EVID))
```

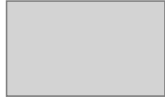
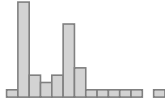
No. of rows in the DataFrame: 2100
 No. of unique study participants: 50
 Total No. of Observations: 1650

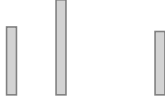
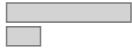
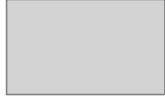
Dataset overview

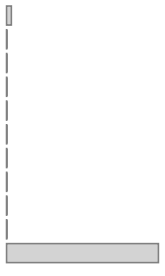
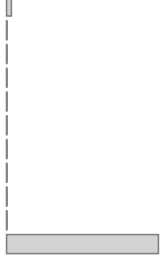
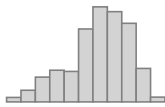
```
tbl = overview_table(prot002)
io = IOBuffer()
show(io, MIME"text/html"(), tbl)
html_output = String(take!(io))
write(joinpath(@__DIR__ , "..", "..", "datasets", "derived",
"poppk_overview.html"), html_output)
tbl
```

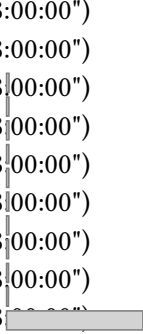
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
1	STUDYID [String15]	1. PROTO-COL-002	2100 (100%)		2100 (100%)	0 (0%)
		1. PROTOCOL-002 (01%)	00276			
		2. PROTOCOL-002 (01%)	00280			
		3. PROTOCOL-002 (01%)	00268			
		4. PROTOCOL-002 (01%)	00279			
		5. PROTOCOL-002 (01%)	00265			
2	USUBJID [String31]	6. PROTOCOL-002 (01%)	00259		2100 (100%)	0 (0%)
		7. PROTOCOL-002 (01%)	00253			
		8. PROTOCOL-002 (01%)	00245			
		9. PROTOCOL-002 (01%)	00264			
		10. PROTOCOL-002 (01%)	00239			
		[40 others]	1680 (80%)			
		1. Dates.Date("2016-08-07")	00280			
		2. Dates.Date("2016-08-13")	00283			
		3. Dates.Date("2016-06-17")	00267			
		4. Dates.Date("2016-06-09")	00269			
		5. Dates.Date("2016-06-13")	00263			
		6. Dates.Date("2016-06-23")	00263			
		7. Dates.Date("2016-06-11")	00261			
		8. Dates.Date("2016-06-19")	00261			
3	RFSTDTC [Date]				2100 (100%)	0 (0%)

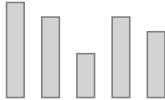
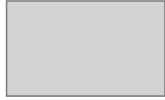
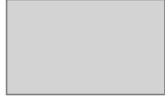
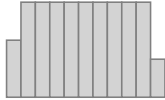
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
4	ARM [String31]	9. Dates.Date("2020-01-23")	100 (4.7%)		2100 (100%)	0 (0%)
		10. Dates.Date("2020-01-31")	820 (41%)			
		[14 others]	798 (38%)			
		1. TEST DRUG 100 mg BID				
5	ACTARM [String31]	2. TEST DRUG 50 mg BID	882 (42%) 630 (30%) 588 (28%)		2100 (100%)	0 (0%)
		3. TEST DRUG 200 mg BID				
		1. TEST DRUG 100 mg BID				
		2. TEST DRUG 50 mg BID	882 (42%) 630 (30%) 588 (28%)			
6	EVID [Int64]	3. TEST DRUG 200 mg BID			2100 (100%)	0 (0%)
		Mean (sd): 0.214 (0.41)				
		min ≤ med ≤ max: 0 ≤ 0 ≤ 1	2 distinct values			
		IQR (CV): 0 (1.92)				
7	NFRLT [Float64]	Mean (sd): 80.7 (90.1)			2100 (100%)	0 (0%)
		min ≤ med ≤ max: 0 ≤ 72 ≤ 384	36 distinct values			
		IQR (CV): 93 (1.12)				
8	NFRLT_BASE [Int64?]	Mean (sd): 42.7 (31.6)			450 (21.4%)	1650 (78.6%)
		min ≤ med ≤	5 distinct values			

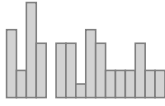
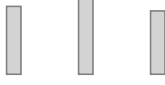
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
9	DOSE_RANK [Int64?]	max: $0 \leq 48 \leq 96$ IQR (CV): 48 (0.74) Mean (sd): 1.44 (0.497) min $\leq$ med $\leq$ max: $1 \leq 1 \leq 2$ IQR (CV): 1 (0.344) Mean (sd): 12 (0)	2 distinct values		450 (21.4%)	1650 (78.6%)
10	HOURS_BETWEEN_DOSES [Int64?]	min $\leq$ med $\leq$ max: $12 \leq 12 \leq 12$ IQR (CV): 0 (0) Mean (sd): 1 (0)	1 distinct values		450 (21.4%)	1650 (78.6%)
11	DOSE_INDEX [Int64?]	min $\leq$ med $\leq$ max: $1 \leq 1 \leq 1$ IQR (CV): 0 (0) Mean (sd): 68.7 (62.2)	1 distinct values		450 (21.4%)	1650 (78.6%)
12	AFRLT [Float64]	min $\leq$ med $\leq$ max: $-0.1 \leq 72 \leq 240$ IQR (CV): 93 (0.905) Mean (sd): 25.3 (36.5)	38 distinct values		2100 (100%)	0 (0%)
13	APRLT [Float64]	min $\leq$ med $\leq$ max: $-0.1 \leq 8 \leq 144$	23 distinct values		2100 (100%)	0 (0%)

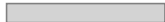
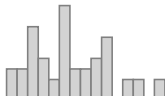
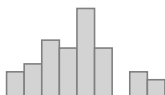
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
14	DOSEA [Int64?]	IQR (CV): 34.5 (1.45) Mean (sd): 113 (58.2) min ≤ med ≤ max: 50 ≤ 100 ≤ 200 IQR (CV): 150 (0.515) Mean (sd): 45.4 (31.7)	3 distinct values		2050 (97.6%)	50 (2.4%)
15	DOSEP [Int64]	min ≤ med ≤ max: 0 ≤ 54 ≤ 81 IQR (CV): 81 (0.699)	3 distinct values		2100 (100%)	0 (0%)
16	PARAMCD [String7]	1. DRUG 2. DOSE Mean (sd): 1 (0)	1650 (78.6%) 450 (21.4%)		2100 (100%)	0 (0%)
17	ALLOQ [Float64?]	min ≤ med ≤ max: 1 ≤ 1 ≤ 1 IQR (CV): 0 (0) Mean (sd): 1.79 (0.41)	1 distinct values		1650 (78.6%)	450 (21.4%)
18	CMT [Int64]	min ≤ med ≤ max: 1 ≤ 2 ≤ 2 IQR (CV): 0 (0.23)	2 distinct values		2100 (100%)	0 (0%)
19	BLQFL [String1]	1. N	2100 (100%)		2100 (100%)	0 (0%)
20	BLQFN [Int64]	Mean (sd): 0 (0) min ≤ med ≤	1 distinct values		2100 (100%)	0 (0%)

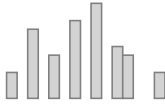
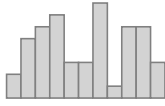
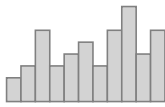
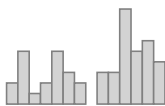
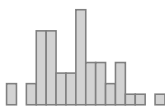
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
21	AMT [Int64?]	max: $0 \leq 0 \leq 0$ IQR (CV): 0 (NaN) Mean (sd): 113 (58.2) min $\leq$ med $\leq$ max: $50 \leq 100 \leq 200$ IQR (CV): 150 (0.515)	3 distinct values		450 (21.4%)	1650 (78.6%)
22	DV [String31?]	1. . 50 (3%) 2. 8.846772558874587 (0.1%) 3. 14.255199934751953 (0.1%) 4. 14.1124533712621083 (0.1%) 5. 9.659734056319570.1%) 6. 32.13362760685766.1%) 7. 18.30016150041246.1%) 8. 8.702756197242799.1%) 9. 37.418368703315204%) 10. 7.607979877873664%) [1541 others] 1582 (95.9%)			1650 (78.6%)	450 (21.4%)
23	AVAL [String31?]	1. . 50 (3%) 2. 8.846772558874587 (0.1%) 3. 14.255199934751953 (0.1%) 4. 14.1124533712621083 (0.1%) 5. 9.659734056319570.1%) 6. 32.13362760685766.1%) 7. 18.30016150041246.1%) 8. 8.702756197242799.1%) 9. 37.418368703315204%) 10. 7.607979877873664%) [1541 others] 1582 (95.9%)			1650 (78.6%)	450 (21.4%)
24	DVL [Float64?]	Mean (sd): 2.7 (1.04) min $\leq$ med $\leq$ max: $-0.448 \leq 2.86 \leq$	1550 distinct values		1600 (76.2%)	500 (23.8%)

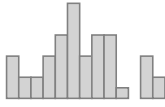
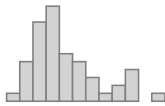
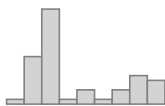
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
25	MDV [Int64]	4.68 IQR (CV): 1.37 (0.384) Mean (sd): 0.214 (0.41) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 0 (1.92)	2 distinct values		2100 (100%)	0 (0%)
26	AVALU [String7?]	1. ng/mL	1650 (100%)		1650 (78.6%)	450 (21.4%)
27	UDTC [DateTime]	1. Dates.DateTime("2020-02-19T08:00:00") 2. Dates.DateTime("2020-02-20T08:00:00") 3. Dates.DateTime("2020-02-16T08:00:00") 4. Dates.DateTime("2020-02-15T08:00:00") 5. Dates.DateTime("2020-02-14T08:00:00") 6. Dates.DateTime("2020-02-18T08:00:00") 7. Dates.DateTime("2020-02-17T08:00:00") 8. Dates.DateTime("2020-02-21T08:00:00") 9. Dates.DateTime("2020-02-10T08:00:00") 10. Dates.DateTime("2020-02-12T08:00:00") [416 others]	1808 (86.1%)		2100 (100%)	0 (0%)
28	II [Int64]	0.214 (0.41) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 0 (1.92)	2 distinct values		2100 (100%)	0 (0%)
29	SS [Int64]	0.214 (0.41) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 0 (1.92)	2 distinct values		2100 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
35	STUDYIDN [Missing]	IQR (CV): 0 (0)			0 (0%)	2100 (100%)
36	SITEID [Int64]	Mean (sd): 2.84 (1.47) min ≤ med ≤ max: 1 ≤ 3 ≤ 5 IQR (CV): 3 (0.519)	5 distinct values		2100 (100%)	0 (0%)
37	SITEIDN [Int64]	Mean (sd): 2 (0) min ≤ med ≤ max: 2 ≤ 2 ≤ 2 IQR (CV): 0 (0)	1 distinct values		2100 (100%)	0 (0%)
38	USUBJIDN [Int64]	Mean (sd): 1 (0) min ≤ med ≤ max: 1 ≤ 1 ≤ 1 IQR (CV): 0 (0)	1 distinct values		2100 (100%)	0 (0%)
39	SUBJID [Int64]	Mean (sd): 262 (14.4) min ≤ med ≤ max: 237 ≤ 262 ≤ 286 IQR (CV): 25 (0.0552)	50 distinct values		2100 (100%)	0 (0%)
40	SUBJIDN [Int64]	Mean (sd): 262 (14.4) min ≤ med ≤ max:	50 distinct values		2100 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
41	AGE [Int64]	237 ≤ 262 ≤ 286 IQR (CV): 25 (0.0552) Mean (sd): 35.8 (9.15) min ≤ med ≤ max: 22 ≤ 34.5 ≤ 53 IQR (CV): 16 (0.256)	27 distinct values		2100 (100%)	0 (0%)
42	SEX [String1]	1. F 2. M Mean (sd): 1.66 (0.474) min ≤ med ≤ max: 1 ≤ 2 ≤ 2 IQR (CV): 1 (0.285)	1386 (66%) 714 (34%)		2100 (100%)	0 (0%)
43	SEXN [Int64]	Mean (sd): 1.66 (0.474) min ≤ med ≤ max: 1 ≤ 2 ≤ 2 IQR (CV): 1 (0.285)	2 distinct values		2100 (100%)	0 (0%)
44	COHORT [Int64]	Mean (sd): 0.98 (0.761) min ≤ med ≤ max: 0 ≤ 1 ≤ 2 IQR (CV): 2 (0.777)	3 distinct values		2100 (100%)	0 (0%)
45	COHORTC [String31]	1. TEST DRUG 100 mg BID 2. TEST DRUG 50 mg BID 3. TEST DRUG 200 mg BID	882 (42%) 630 (30%) 588 (28%)		2100 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
46	ROUTE [String7]	1. ORAL	2100 (100%)		2100 (100%)	0 (0%)
47	ROUTEN [Missing]				0 (0%)	2100 (100%)
48	FORM [String7]	1. TABLET	2100 (100%)		2100 (100%)	0 (0%)
49	FORMN [Int64]	Mean (sd): 4 (0) min ≤ med ≤ max: 4 ≤ 4 ≤ 4 IQR (CV): 0 (0)	1 distinct values		2100 (100%)	0 (0%)
50	COUNTRY [String15]	1. GERMANY 2. UNITED STATES 3. JAPAN	756 (36%) 714 (34%) 630 (30%)		2100 (100%)	0 (0%)
51	country_name [Missing]				0 (0%)	2100 (100%)
52	country_number [Missing]				0 (0%)	2100 (100%)
53	HTBL [Float64]	Mean (sd): 168 (7.29) min ≤ med ≤ max: 156 ≤ 166 ≤ 185 IQR (CV): 12 (0.0435)	24 distinct values		2100 (100%)	0 (0%)
54	WTBL [Float64]	Mean (sd): 70 (10.3) min ≤ med ≤ max: 51.5 ≤ 70.2 ≤ 91.6 IQR (CV): 13.6 (0.147)	46 distinct values		2100 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
55	CREATBL [Float64]	Mean (sd): 1.14 (0.192) min ≤ med ≤ max: 0.8 ≤ 1.15 ≤ 1.5 IQR (CV): 0.3 (0.168)	8 distinct values		2100 (100%)	0 (0%)
56	TBILBL [Float64]	Mean (sd): 1 (0.301) min ≤ med ≤ max: 0.5 ≤ 1 ≤ 1.5 IQR (CV): 0.6 (0.3)	11 distinct values		2100 (100%)	0 (0%)
57	ASTBL [Float64]	Mean (sd): 26 (5.95) min ≤ med ≤ max: 15 ≤ 26.5 ≤ 35 IQR (CV): 10 (0.229)	20 distinct values		2100 (100%)	0 (0%)
58	ALTBL [Float64]	Mean (sd): 29.8 (8.04) min ≤ med ≤ max: 15 ≤ 33 ≤ 40 IQR (CV): 14 (0.27)	21 distinct values		2100 (100%)	0 (0%)
59	BMIBL [Float64]	Mean (sd): 24.9 (3.25) min ≤ med ≤ max: 18.1 ≤ 25.1 ≤ 33.2 IQR (CV): 4.84 (0.131)	50 distinct values		2100 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
60	BSABL [Float64]	Mean (sd): 1.8 (0.155) min ≤ med ≤ max: 1.51 ≤ 1.79 ≤ 2.13 IQR (CV): 0.194 (0.086)	49 distinct values		2100 (100%)	0 (0%)
61	CRCLBL [Float64]	Mean (sd): 7434 (2513) min ≤ med ≤ max: 3739 ≤ 6669 ≤ 14261 IQR (CV): 2764 (0.338)	50 distinct values		2100 (100%)	0 (0%)
62	EGFRBL [Float64]	Mean (sd): 495 (124) min ≤ med ≤ max: 346 ≤ 428 ≤ 736 IQR (CV): 212 (0.25)	49 distinct values		2100 (100%)	0 (0%)
63	RECSEQ [String]	1. Base.EachStringIndex{String} ("PROTOCOL-002-001-00250") 42 (2%) 2. Base.EachStringIndex{String} ("PROTOCOL-002-001-00264") 42 (2%) 3. Base.EachStringIndex{String} ("PROTOCOL-002-001-00271") 42 (2%) 4. Base.EachStringIndex{String} ("PROTOCOL-002-001-00283") 42 (2%) 5. Base.EachStringIndex{String} ("PROTOCOL-002-001-00279") 42 (2%) 6. Base.EachStringIndex{String} ("PROTOCOL-002-001-00284") 1680 (80%) 7. Base.EachStringIndex{String} ("PROTOCOL-002-001-00254")			2100 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
		8. Base.EachStringIndex{String}				
		("PROTOCOL-002-001-00285")				
		9. Base.EachStringIndex{String}				
		("PROTOCOL-002-001-00247")				
		10. Base.EachStringIndex{String}				
		("PROTOCOL-002-001-00281")				
		[40 others]				

Dimensions: 2100 x 63

Duplicate rows: 0

Tabular Summary of Subject Characteristics

```
adata_subj = unique(prot002, :USUBJID) # Subset to one row per subject
(USUBJID)
table_one_mod( # bespoke function table_one_mod - renders Total column in the
last. Use table_one to get it in the front.
adata_subj,
[
:SEX => "Sex",
:AGE => "Age [yr]",
:WTBL => "Baseline Weight [kg]",
:BMIBL => "Baseline BMI [kg/m²]",
:EGFRBL => "Baseline eGFR (CKD-2021) [mL/min/1.73 m²]"
],
groupby=:STUDYID => "Study",
show_n=true,
footnotes=[
"Abbreviations: BMI = body mass index, CKD = chronic kidney disease,
eGFR = estimated glomerular filtration rate, max = maximum, min =
minimum, SD = Standard Deviation",
>Note: Missing row appears only when there are missing values"
]
)
```

	Study	
	PROTOCOL-002 (n=50)	Total (n=50)
Sex		
F	33 (66%)	33 (66%)
M	17 (34%)	17 (34%)

	Study	
	PROTOCOL-002 (n=50)	Total (n=50)
Age [yr]		
Mean (SD)	35.8 (9.24)	35.8 (9.24)
Median [Min, Max]	34.5 [22, 53]	34.5 [22, 53]
Baseline Weight [kg]		
Mean (SD)	70 (10.4)	70 (10.4)
Median [Min, Max]	70.2 [51.5, 91.6]	70.2 [51.5, 91.6]
Baseline BMI [kg/m²]		
Mean (SD)	24.9 (3.28)	24.9 (3.28)
Median [Min, Max]	25.1 [18.1, 33.2]	25.1 [18.1, 33.2]
Baseline eGFR (CKD-2021) [mL/min/1.73 m²]		
Mean (SD)	495 (125)	495 (125)
Median [Min, Max]	428 [346, 736]	428 [346, 736]

Abbreviations: BMI = body mass index, CKD = chronic kidney disease,

eGFR = estimated glomerular filtration rate, max = maximum, min = minimum, SD = Standard Deviation

Note: Missing row appears only when there are missing values

Correlation between covariates

```
pair_plot = @chain prot002 begin
  @select :STUDYIDN :USUBJID :AGE :BMIBL :EGFRBL :WTBL # select only
  covariate data & unique identifiers
  unique # filter to keep only unique
  @select :AGE :BMIBL :EGFRBL :WTBL
  @rename begin # Renaming the columns
    $("Age [yr]") = :AGE
    $(":BMI [kg/m²]") = :BMIBL
    $(":eGFR [mL/min/m²]") = :EGFRBL
    $(":Weight [kg]") = :WTBL
  end
  dropmissing

  # The following function is defined in 20-poppk/julia-functions/pairplot-
  with-regression.
  # Regression Line with CI Band is not available in PairPlots.jl
  pairplot_with_regression(
    color=:royalblue3,
    title="Correlation between Continuous-Valued Covariates",
```



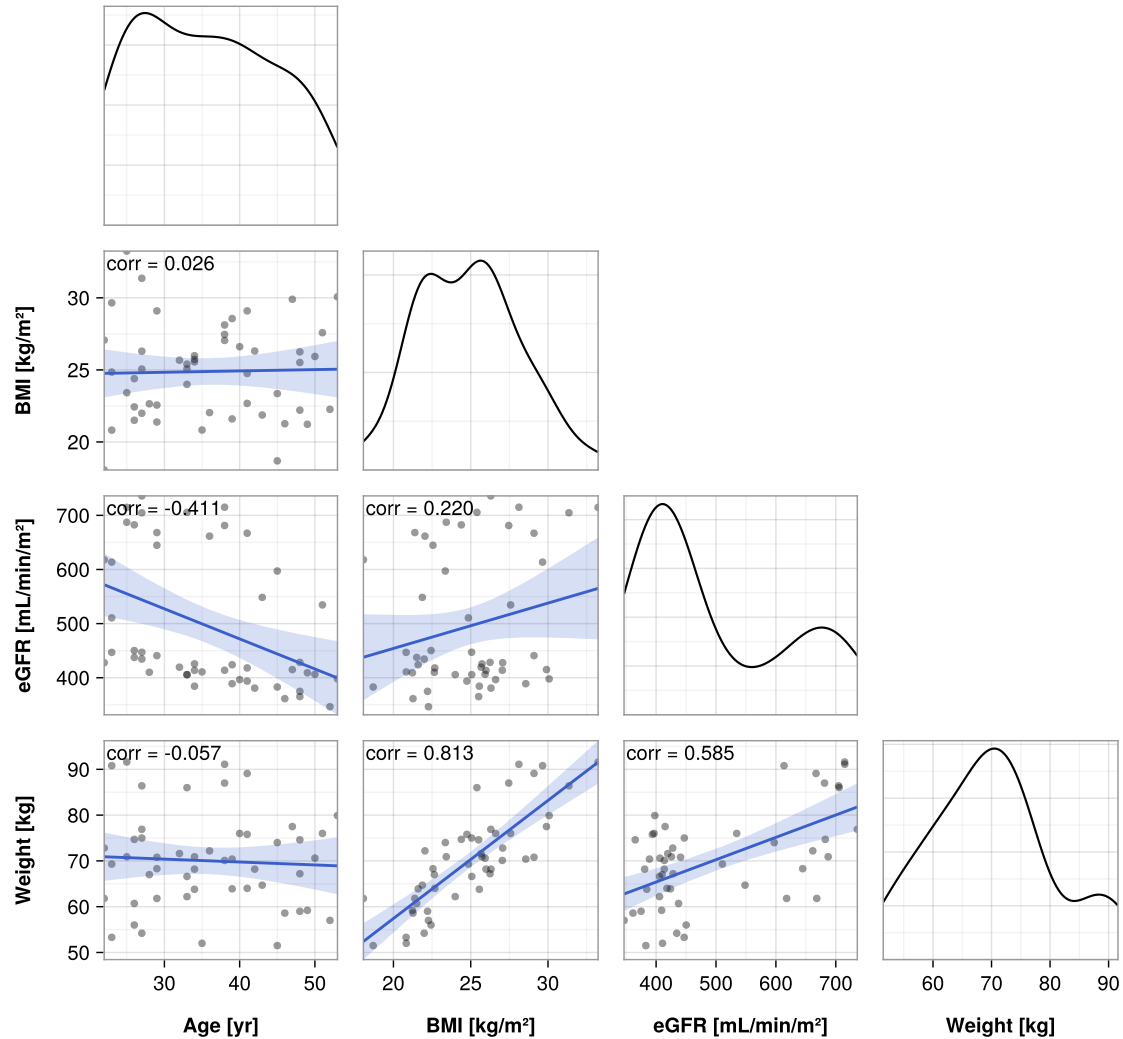
```

    subtitle="CDISCILOT01",
    size=(800, 800)
)
end

```

Correlation between Continuous-Valued Covariates

CDISCILOT01



Visualize Covariate Distributions

Continuous variates

```

dist_layer = @chain prot002 begin
  @select :STUDYID :USUBJID :AGE :BMIBL :EGFRBL :WTBL # select only
  continuous covariate data & unique identifiers
end

```

```

unique # filter to keep only unique
@rename begin # Renaming the columns
  $("Age [yr]") = :AGE
  $("BMI [kg/m²]") = :BMIBL
  $("eGFR [mL/min/m²]") = :EGFRBL
  $("Weight [kg]") = :WTBL
end
stack(_, [:"Age [yr]", : "BMI [kg/m²]", : "eGFR [mL/min/m²]", : "Weight
[kg]"])
# dropmissing

data(_) *
mapping(
  :value,
  layout=:variable,
  color=:variable,
)

end

```

```

Layer
  transformation: identity
  data: AlgebraOfGraphics.Columns{DataFrames.DataFrameColumns{DataFrame}}
  positional:
    1: value
  named:
    layout: variable
    color: variable

```

Density Plot

```

density_plot = dist_layer *
  AlgebraOfGraphics.density(datalimits=extrema)

density_plot = draw(
  density_plot,
  figure=(
    title="Density Plots of Continuous Covariates",
    subtitle="CDISCIPL0T01",
    size=(800, 600)
  ),
  axis=(;
    xlabel="",
  ),
  facet=(
    linkyaxes=:none,

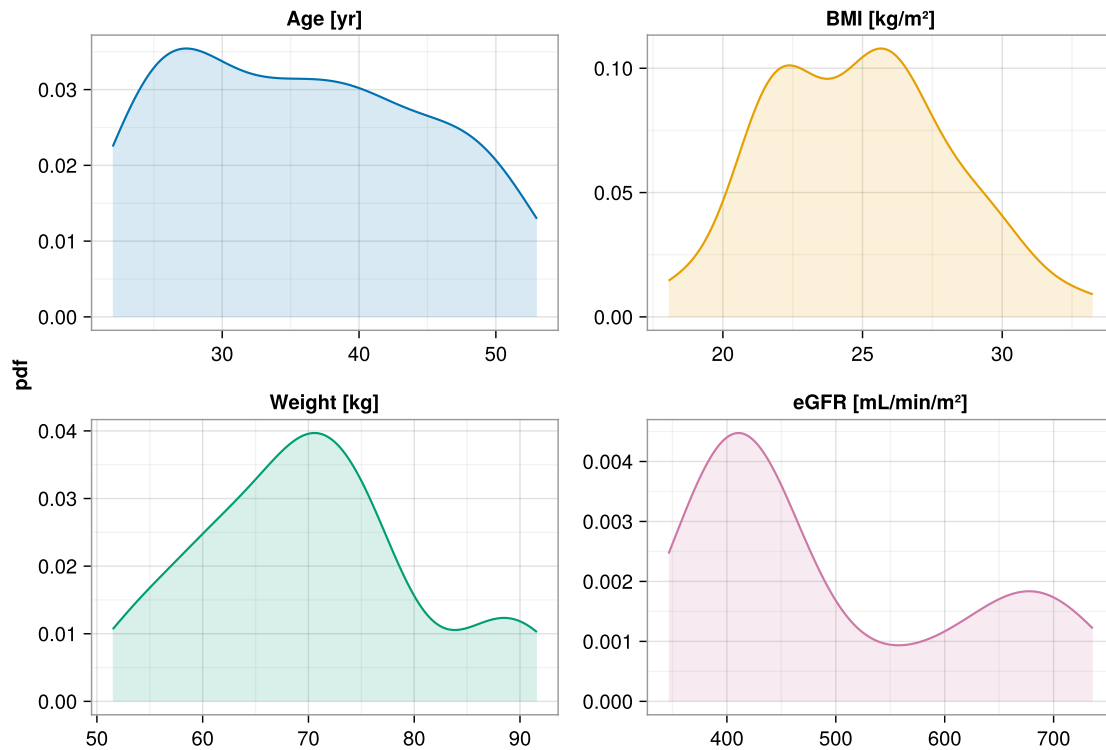
```

```

    linkxaxes=:none,
  ),
  legend=(show=false,)
)

```

Density Plots of Continuous Covariates
CDISCPILOT01



```

hist_plot = dist_layer *
    AlgebraOfGraphics.histogram(bins=15, datalimits=extrema)

hist_plot = draw(
    hist_plot,
    figure=(
        title="Histograms of Continuous Covariates",
        subtitle="CDISCPILOT01",
        size=(800, 600)
    ),
    axis=(;
        xlabel="",
    ),
    facet=(
        linkxaxes=:none,

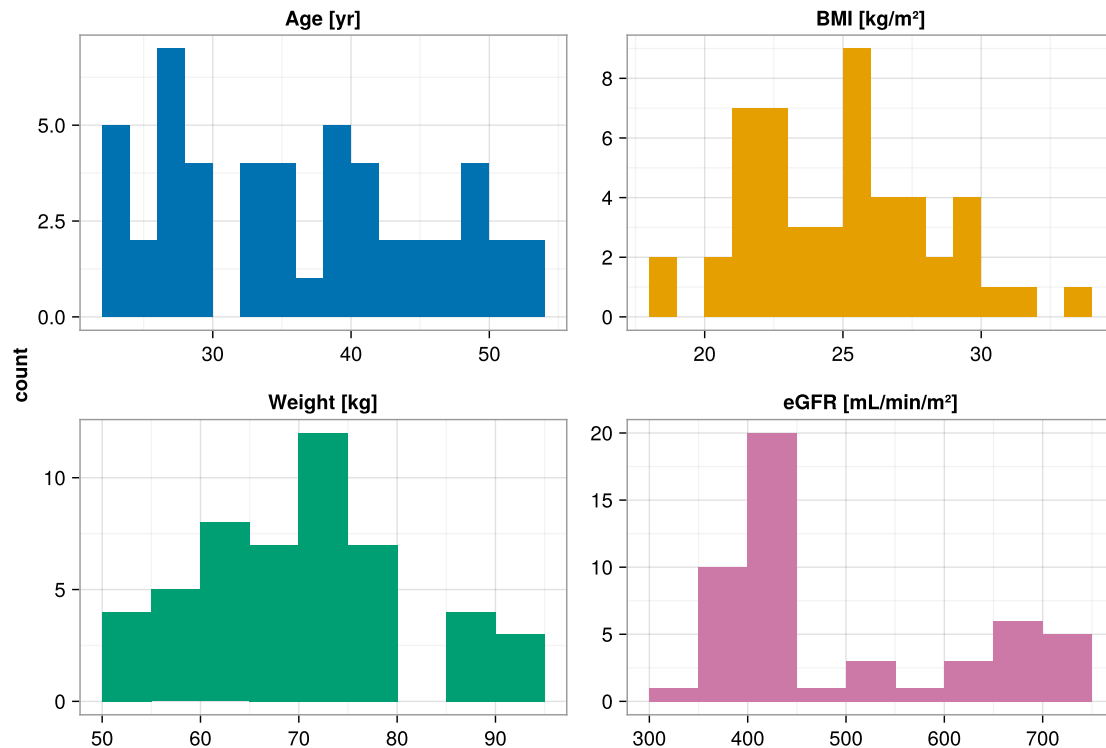
```

```

    linkxaxes=:none,
  ),
  legend=(show=false,)
)

```

Histograms of Continuous Covariates
CDISCPilot01



Categorical variates

```

cat_cov = @chain prot002 begin
  @select :USUBJID :SEX # Select Categorical covariates
  unique # Getting unique subjects - onw row per subject
  @groupby :SEX
  @combine :n = length(:USUBJID)
  dropmissing

  data(_) *
  mapping(
    :SEX,
    :n,
    color=:SEX
  )
end

```

```

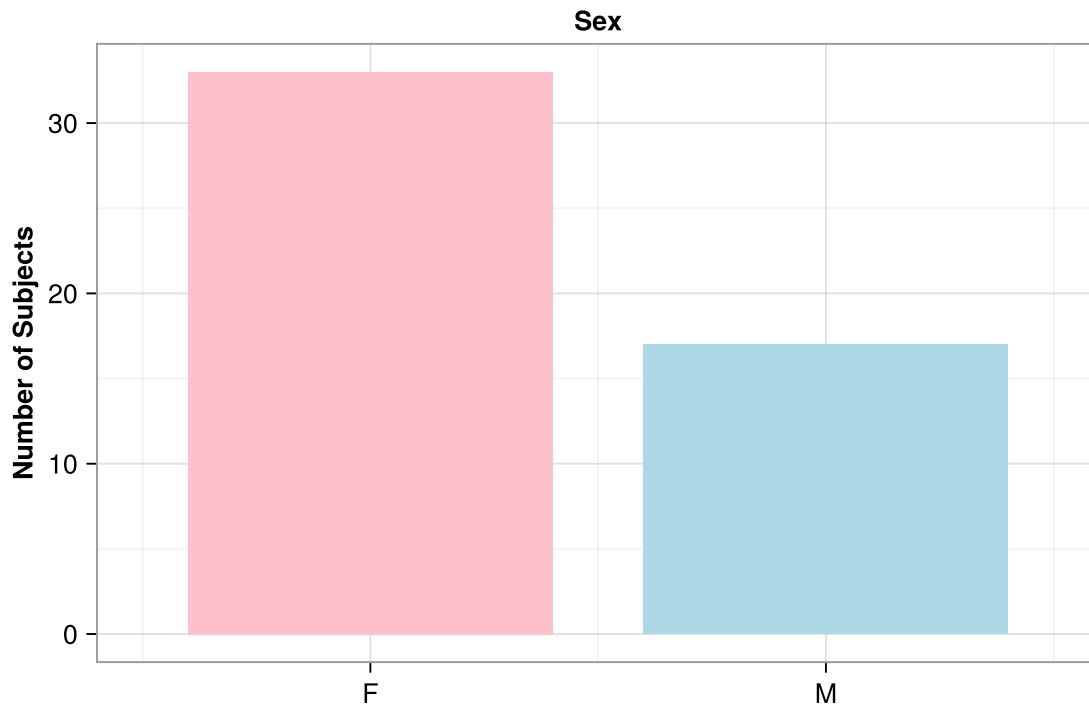
    ) *
    visual(BarPlot,
          color=:lightblue
    )
end

draw(cat_cov,
     figure=(;
         title="Barplots of Categorical Covariates",
         subtitle="CDISCIPL0T01"
     ),
     axis=(;
         title="Sex",
         yticks=0:10:50,
         ylabel="Number of Subjects",
         xlabel=""
     ),
     scales(;
         Color=(;
             palette=[:pink, :lightblue]
         )
     ),
     legend=(;
         show=false
     )
)

```

Barplots of Categorical Covariates

CDISCPLOT01



Tabular Summary of assigned treatments

```
adata_subj = unique(prot002, :USUBJID) # Subset to one row per subject
(USUBJID)
adata_subj.TX = categorical(adata_subj.COHORTC,
levels=unique(adata_subj.COHORTC))

table_one_mod( # bespoke function table_one_mod - renders Total column in the
last. Use table_one to get it in the front.
  adata_subj,
  [
    :ARM => stats_cat => "Actual Treatment, n (%)"
  ],
  groupby=:STUDYID => "Study",
  show_n=true,
  # sort = true
  # footnotes=["Abbreviations: BMI = body mass index, CKD = chronic kidney
disease, eGFR = estimated glomerular filtration rate, max = maximum, min =
minimum, SD = Standard Deviation"]
)
```

	Study	
	PROTOCOL-002 (n=50)	Total (n=50)
Actual Treatment, n (%)		
TEST DRUG 100 mg BID	21 (42.0%)	21 (42.0%)
TEST DRUG 200 mg BID	14 (28.0%)	14 (28.0%)
TEST DRUG 50 mg BID	15 (30.0%)	15 (30.0%)

Tabular Summary of BLOQ by Nominal Time relative to previous dose

```

blq_smr = @chain prot002 begin
  @transform(:NFRLTC = @. string(round(:NFRLT, digits = 2)))
  @rsubset :EVID == 0
  @groupby :NFRLTC
  @combine begin
    :N = length(:DV)
    :NBLQ = sum(:BLQFN)
    :BLQ [%] = round(sum(:BLQFN) / length(:DV) * 100, digits = 1)
  end
end

total_n = sum(blq_smr.N)
total_nblq = sum(blq_smr.NBLQ)
total_pc_nblq = round(sum(blq_smr.NBLQ) / total_n * 100, digits=1)

# Formatting the TOTAL row to bold and making a DataFrame
blq_tot_smr = DataFrame([
  :NFRLTC => [styled"{bold:Total}"],
  :N => [styled"{bold:$total_n}"],
  :NBLQ => [styled"{bold:$total_nblq}"],
  Symbol("BLQ [%]") => [styled"{bold:$total_pc_nblq}"]
])

@chain vcat(blq_smr, blq_tot_smr) begin
  @rtransform begin
    :NBLQ = :NBLQ == 0 ? "-" : :NBLQ
    :BLQ [%] = :BLQ [%] == 0 ? "-" : :BLQ [%]
  end

  simple_table(
    [
      :NFRLTC => Annotated("Nominal Sampling Time [hr]", "Relative to
previous dose", label="a"),
      :N => Annotated("N", "Number of Samples", label="b"),
      :NBLQ => Annotated("NBLQ", "Number of BLQ samples", label="c"),

```

```

        Symbol("BLQ [%]")
    ],
    halign=[:left, :right, :right, :right]
)
end

```

Nominal Sampling Time [hr] ^a	N ^b	NBLQ ^c	BLQ [%]
0.0	50	-	-
0.25	50	-	-
0.5	50	-	-
0.75	50	-	-
1.0	50	-	-
1.5	50	-	-
2.0	50	-	-
3.0	50	-	-
4.0	50	-	-
6.0	50	-	-
8.0	50	-	-
12.0	50	-	-
24.0	50	-	-
48.0	50	-	-
72.0	50	-	-
96.0	50	-	-
96.25	50	-	-
96.5	50	-	-
96.75	50	-	-
97.0	50	-	-
97.5	50	-	-
98.0	50	-	-
99.0	50	-	-
100.0	50	-	-
102.0	50	-	-
104.0	50	-	-

Nominal Sampling Time [hr] ^a	N ^b	NBLQ ^c	BLQ [%]
108.0	50	-	-
144.0	50	-	-
192.0	50	-	-
240.0	50	-	-
288.0	50	-	-
336.0	50	-	-
384.0	50	-	-
Total	1650	0	0.0

^aRelative to previous dose

^bNumber of Samples

^cNumber of BLQ samples

Histograms of samples per subject

```

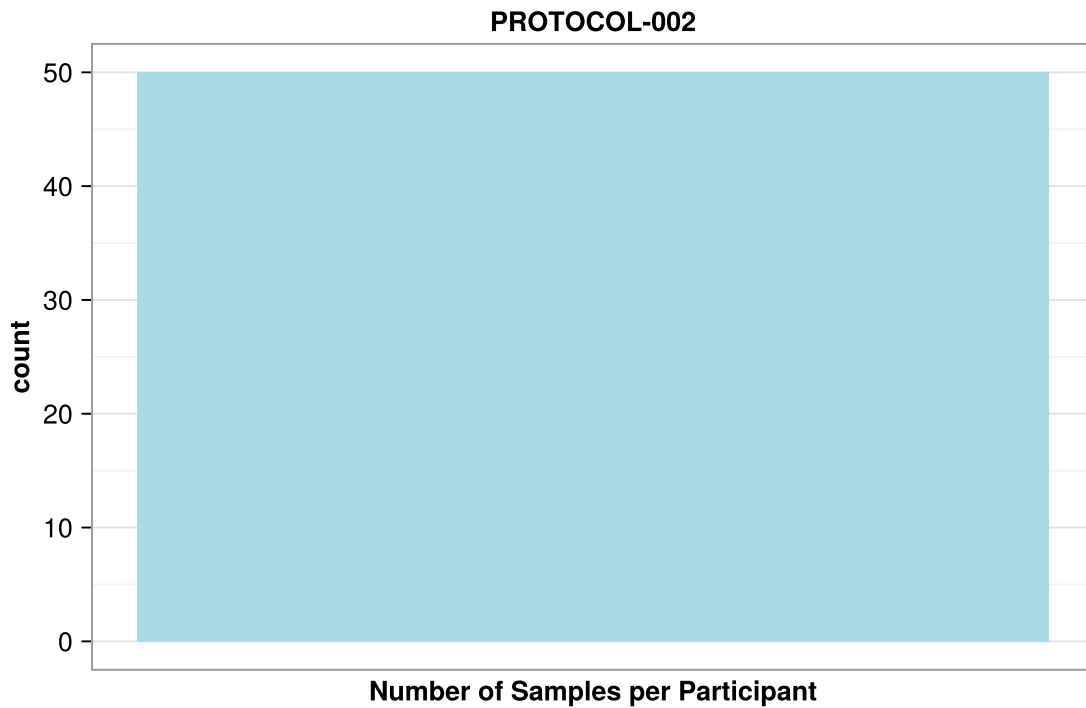
samples_hist = @chain prot002 begin
  # @rsubset :EVID == 0
  @groupby :USUBJID :STUDYID
  @combine :NOBS = length(:DV)
  dropmissing

  data(_) *
  mapping(
    :NOBS,
    layout=:STUDYID
  ) *
  AlgebraOfGraphics.histogram(bins=25) *
  visual(color=:lightblue)
end

draw(
  samples_hist,
  figure=(
    title="Histogram of the Number of PK samples, by study",
    subtitle="CDISCPILOT01"
  ),
  axis=(
    yticks=0:10:100,
    xticks=0:5:100,
    xlabel="Number of Samples per Participant"
  )
)

```

Histogram of the Number of PK samples, by study CDISCPILOT01



Histograms of Sampling Times

Relative to previous dose

```
dose_freq_lookup = DataFrame(
    CDISC_VALUE = ["QD", "BID", "TID", "QID"],
    HOURS_BETWEEN_DOSES = [24, 12, 8, 6]
)
```

```
4x2 DataFrame
  Row | CDISC_VALUE  HOURS_BETWEEN_DOSES
     | String      Int64
-----|-----
  1  | QD           24
  2  | BID          12
  3  | TID           8
  4  | QID           6
```

```
prot002 = @chain prot002 begin
    leftjoin(
```

```

    dose_freq_lookup,
    on = :HOURS_BETWEEN_DOSES,
    matchmissing = :equal
  )
  rename(:CDISC_VALUE => :FREQC)
  # @orderby([:STUDYID, :USUBJID, :EVID])

end
first(prot002,5)

```

```

5x64 DataFrame
  Row | STUDYID      USUBJID      RFSTDTC      ARM
...   | String15     String31      Date       String31
...   |
1 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
...
2 | PROTOCOL-002 PROTOCOL-002-001-00238 2020-02-11 TEST DRUG 200 mg BID
3 | PROTOCOL-002 PROTOCOL-002-001-00239 2020-02-17 TEST DRUG 200 mg BID
4 | PROTOCOL-002 PROTOCOL-002-001-00240 2020-02-07 TEST DRUG 50 mg BID
5 | PROTOCOL-002 PROTOCOL-002-001-00241 2020-02-17 TEST DRUG 100 mg BID
...
60 columns
omitted

```

```

prot002 = sort(prot002,[:STUDYID, :USUBJID, :EVID])
first(prot002,5)

```

```

5x64 DataFrame
  Row | STUDYID      USUBJID      RFSTDTC      ARM
...   | String15     String31      Date       String31
...   |
1 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
...
2 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
3 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
4 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
5 | PROTOCOL-002 PROTOCOL-002-001-00237 2020-02-06 TEST DRUG 50 mg BID
...
60 columns
omitted

```

```

dist_apr1t = @chain prot002 begin
  @rsubset begin
    :EVID == 0
    # :BLQFL == 0
  end
  # dropmissing

  data(_) *
  mapping(
    :APRLT,
    # row=:FREQC => sorter(["QD", "BID", "TID", "QID", missing])
  ) *
  AlgebraOfGraphics.histogram(bins=35, datalimits=extrema) *
  visual(color=:lightblue)

  draw(
    _,
    figure=(
      title="Histogram of Actual Time from First Dose, by Dosing
Frequency",
      subtitle="CDISCP1LOT01",
      size=(800, 600)
    ),
    axis=(;
      # xticks = IntervalsBetween(3)
    ),
    facet=(
      linkyaxes=:none,
      linkxaxes=:none
    )
  )
end

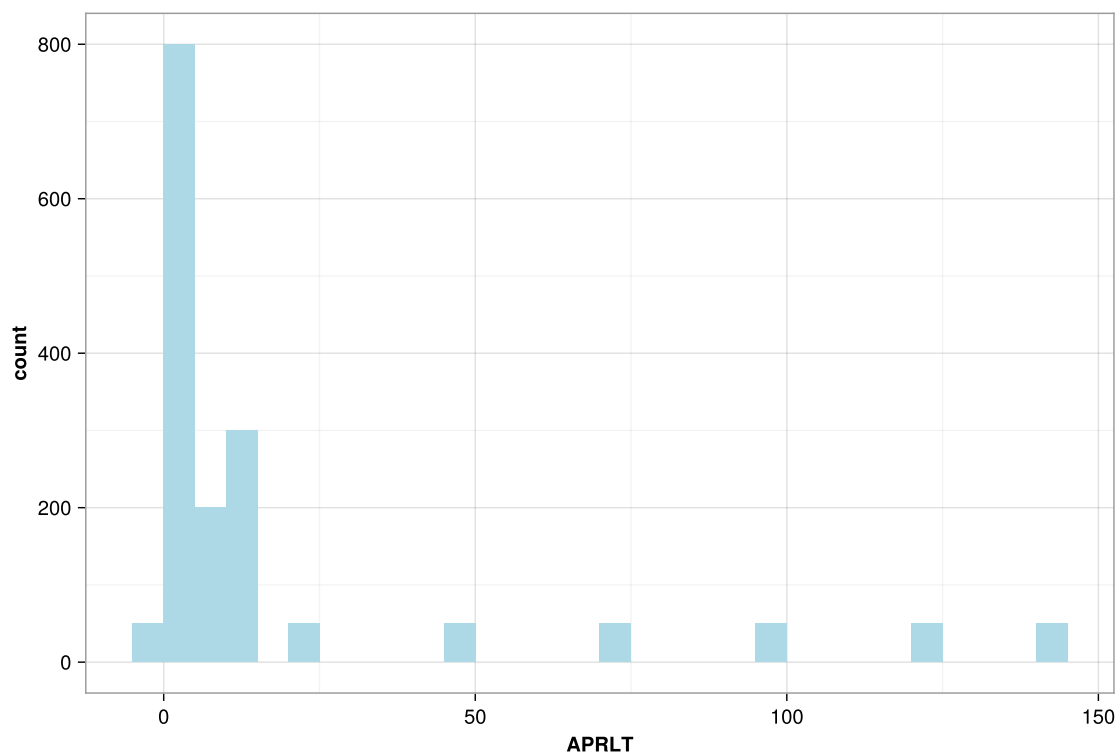
# axes = dist_apr1t.figure.content

# axes[1, 1].xticks = 0:6:100
# axes[2, 1].xticks = 0:2:100
# axes[3, 1].xticks = 0:1:100

dist_apr1t

```

Histogram of Actual Time from First Dose, by Dosing Frequency
CDISCPLOT01



Relative to First Dose

```
dist_nfrlt=@chain prot002 begin
  @transform :NFR LTD = :NFR LT / 24
  @subset begin
    :EVID == 0
    # :BLQFL == 0
  end

  data(_) *
  mapping(
    :NFR LTD,
    # row=:FREQC => sorter(["Single Dose", "QD", "BID"])
  ) *
  AlgebraOfGraphics.histogram(bins=35, datalimits=extrema,
normalization=:none) *
  visual(color=:lightblue)

  draw(,
    figure=(
      title="Histogram of Nominal Time from First Dose, by Dosing
```

```

Frequency",
    subtitle="CDISCPIL0T01",
    size=(800, 600)
),
axis=(;
# xticks = 0:2:100,
xlabel = "Nominal Time Relative to First Dose [d]"
),
facet=(
    linkyaxes=:none,
    linkxaxes=:none
)
)
end

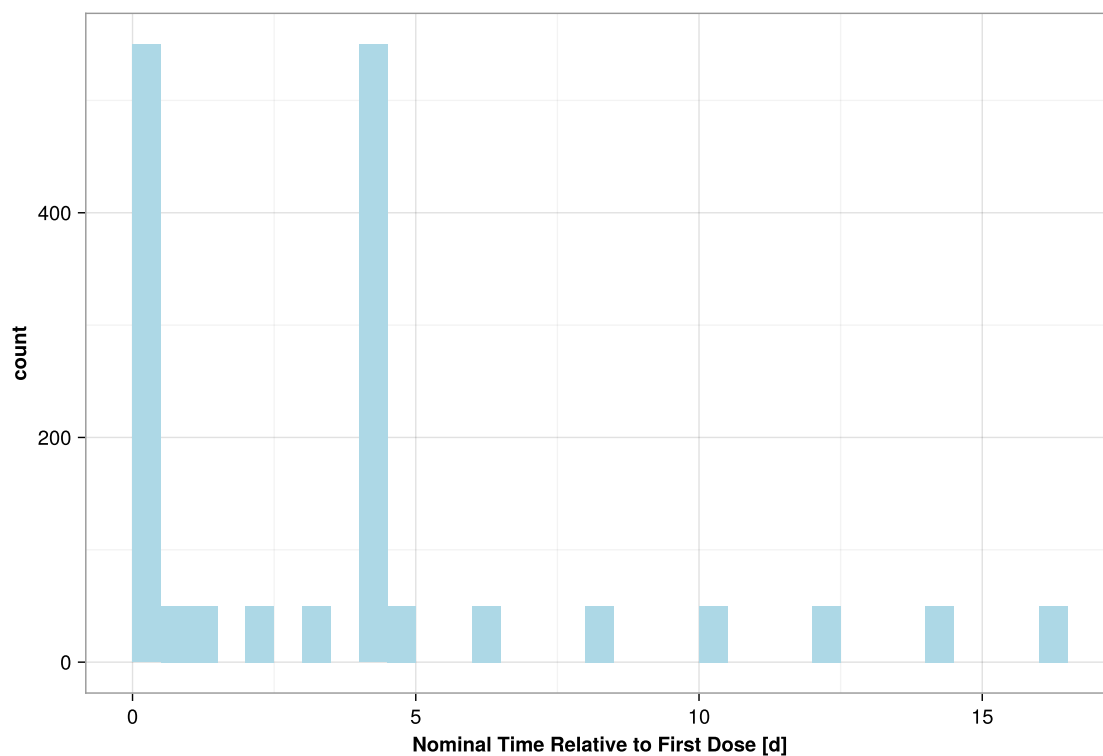
# axes = dist_nfrlt.figure.content

# axes[1, 1].xticks = 0:0.5:100
# axes[2, 1].xticks = 0:2:100
# axes[3, 1].xticks = 0:2:100

dist_nfrlt

```

Histogram of Nominal Time from First Dose, by Dosing Frequency
CDISCPLOT01



Derive Plotting Data

```
pdata = @chain prot002 begin
  @rsubset :STUDYID == "PROTOCOL-002"

  @transform :TXA = categorical(
    ismissing(:FREQC) ? missing :
    replace(:FREQC, r" \(\BID\) " => "BID")
  )

  @rtransform :DVNUM =
    ismissing(:DV) ? missing : tryparse(Float64, :DV)

  @rtransform :DNDV =
    (ismissing(:DVNUM) || ismissing(:DOSEA) || :DOSEA == 0) ?
    missing :
    :DVNUM / :DOSEA

  @select :USUBJID :NFRLT :AFRLT :EVID :AMT :DV :DVNUM :DNDV :BLQFL :FREQC :TXA :DOSEA
end
```

```

2100x12 DataFrame
  Row | USUBJID          NFRLT   AFRLT   EVID   AMT   DV
...   |          String31   Float64  Float64  Int64  Int64? String31?
...   |
1 | PROTOCOL-002-001-00237  0.0    -0.1     0 missing .
...
2 | PROTOCOL-002-001-00237  0.25   0.25     0 missing 1.2785519997
3 | PROTOCOL-002-001-00237  0.5    0.5      0 missing 1.9297970554
4 | PROTOCOL-002-001-00237  0.75   0.75     0 missing 4.1741064106
5 | PROTOCOL-002-001-00237  1.0    1.0      0 missing 6.3593990951
...
6 | PROTOCOL-002-001-00237  1.5    1.5      0 missing 8.0486565556
7 | PROTOCOL-002-001-00237  2.0    2.0      0 missing 5.9708062394
8 | PROTOCOL-002-001-00237  3.0    3.0      0 missing 11.283664390
: | :                :      :      :      :      :
2094 | PROTOCOL-002-001-00286  24.0   24.0     1      50 missing
...
2095 | PROTOCOL-002-001-00286  36.0   36.0     1      50 missing
2096 | PROTOCOL-002-001-00286  48.0   48.0     1      50 missing
2097 | PROTOCOL-002-001-00286  60.0   60.0     1      50 missing
2098 | PROTOCOL-002-001-00286  72.0   72.0     1      50 missing
...
2099 | PROTOCOL-002-001-00286  84.0   84.0     1      50 missing
2100 | PROTOCOL-002-001-00286  96.0   96.0     1      50 missing
7 columns and 2085 rows
omitted

```

```

safe_median(x) = begin
  v = collect(skipmissing(x))
  v = filter(!isnothing, v)
  isempty(v) ? missing : median(v)
end

pdata_smr = @chain pdata begin
  @rsubset :EVID == 0
  @groupby :TXA :NFRLT
  @combine(
    :DVP50 = safe_median(:DVNUM),
    :DNDVP50 = safe_median(:DNDV),
  )
end

```

```

33x4 DataFrame
  Row | TXA      NFRLT   DVP50      DNDVP50
      | Cat...?  Float64  Float64?   Float64?

```


1	missing	0.0	missing	missing
2	missing	0.25	2.76712	0.0294587
3	missing	0.5	4.17659	0.0444638
4	missing	0.75	9.03386	0.0961742
5	missing	1.0	13.7634	0.146525
6	missing	1.5	17.4194	0.185446
7	missing	2.0	12.9224	0.137571
8	missing	3.0	24.4208	0.259983
:	:	:	:	:
27	missing	108.0	43.564	0.463781
28	missing	144.0	16.4672	0.175309
29	missing	192.0	3.15875	0.033628
30	missing	240.0	3.51457	0.037416
31	missing	288.0	4.12581	0.0439232
32	missing	336.0	4.48484	0.0477454
33	missing	384.0	1.46621	0.0156092

18 rows omitted

Defining plot layers

```
individual_lines =
  data(pdata) *
  mapping(
    :AFRLT,
    :DV,
    group=:USUBJID
  ) *
  visual(
    Lines,
    alpha = 0.5,
    color=:gray75
  )

lloq_line =
  mapping(0) * # LLQ = 5
  visual(
    HLines,
    linestyle=:dash,
    color=:red,
    linewidth=1,
    label="LLQ (0 ng/mL)"
  )

rug_nfrlt =
  mapping(unique(pdata_smr.NFRLT)) *
  visual(VLines; color=:red, ymin=0, ymax=0.05, linewidth=1)
```

```

median_layer =
  data(pdata_smr) *
  mapping(
    :NFRLT,
    :DVP50
  ) *
  visual(Lines)

dn_median_layer =
  data(pdata_smr) *
  mapping(
    :NFRLT,
    :DNDVP50
  ) *
  visual(Lines)

median_grouping =
  mapping(
    color=:TXA => sorter(unique(pdata_smr.TXA)),
    linestyle=:TXA => sorter(unique(pdata_smr.TXA))
  )

layout_by_TXA =
  mapping(layout=:TXA => sorter(unique(pdata_smr.TXA)))

```

```

Layer
  transformation: identity
  data: Nothing
  positional:
  named:
    layout: :TXA => AlgebraOfGraphics.Renamer{CategoricalVector{Union{Missing,
String}}, UInt32, String, CategoricalValue{String, UInt32}, Missing},
Vector{String}}(Union{Missing, CategoricalValue{String, UInt32}}[missing],
["missing"])

```

Median C-T Profiles

```

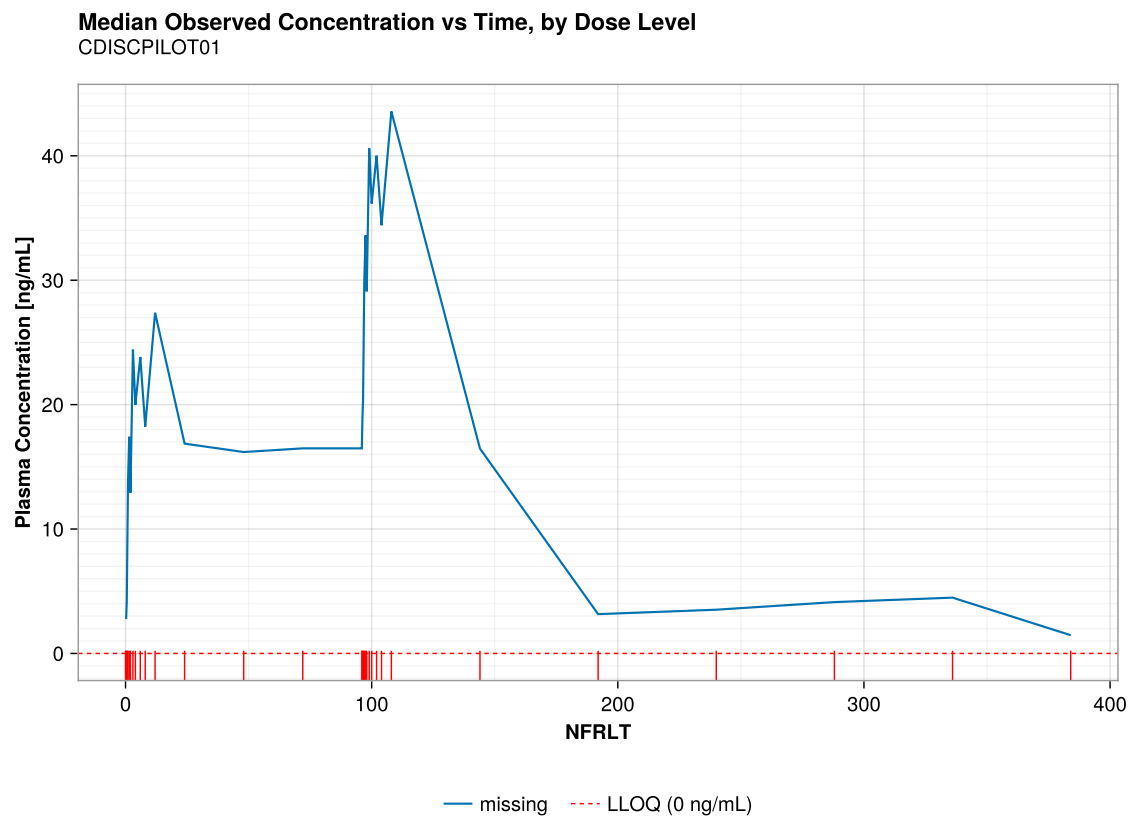
with_theme(my_log_theme) do
  median_ct_plot =
    (
      median_layer * median_grouping
      +
      lloq_line
      +
      rug_nfrlt
    ) |>

```

```

draw(
  figure=(
    title="Median Observed Concentration vs Time, by Dose Level",
    subtitle="CDISCPIL0T01",
    size=(800, 600)
  ),
  axis=(ylabel="Plasma Concentration [ng/mL]",
    limits=(nothing, nothing, nothing, nothing)
  ),
  legend=(
    position=:bottom,
    nbanks=2,
    framevisible=false,
    titlesize=0
  )
)
end

```



Dose Normalized Median C-T Profiles

```
with_theme(my_log_theme) do
  dn_median_ct_plot =
    (
      dn_median_layer * median_grouping
      +
      rug_nfrlt
    ) |> draw(
      figure=(
        title="Median Dose-Normalized Observed Concentration vs Time,
by Dose Level",
        subtitle="CDISCPILOT01",
        size=(800, 600)
      ),
      axis=(
        ylabel="Plasma Concentration [ng/mL]",
        limits=(nothing, nothing, nothing, nothing)
      ),
      legend=(
        position=:bottom,
        nbanks=2,
        framevisible=false,
        titlesize=0
      )
    )
end
```

Median Dose-Normalized Observed Concentration vs Time, by Dose Level
CDISPILOT01

